

Hariprashad Ravikumar

PhD Candidate specializing in High-Performance Computing, Deep Learning, and AI for Science

Expertise in GPU-accelerated computing with C++/CUDA and Python

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Experience

Graduate Research Assistant, New Mexico State University

(Aug 2021 - Present)

PhD Project: Lattice Quantum Chromodynamics & Machine Learning Approaches

- Generated 30,000+ high-fidelity synthetic data points by solving Partial Differential Equations with large-scale Hybrid Monte Carlo simulations (Markov Chain stochastic process) and built an end-to-end AI for Science pipeline to model the underlying physics, achieving over 98% predictive accuracy with symbolic regression machine learning.
- Developed GPU-accelerated CUDA/C++ pipelines for large-scale FFT-based feature extraction, reducing data processing time by 10x
- Developed production-grade Python, Lua, & Mathematica packages for reproducible statistical analysis, ensuring numerical stability in multi-stage workflows (jackknife resampling, uncertainty propagation)

Independent Collaborations

1. **Los Alamos National Laboratory** - Lattice Quantum Chromodynamics

(May 2024 - Present)

- Accelerated multi-terabyte scientific calculations by developing and optimizing parallelized C++ CUDA kernels for GPU-accelerated HPC clusters (NERSC Perlmutter), significantly reducing runtime for large-scale Hamiltonian Monte Carlo simulations.
- Managed and executed 75,000+ CPU/GPU compute hours by designing and deploying custom SLURM workflows for large-scale job orchestration
- Investigated advanced simulation techniques using gradient flow, a method conceptually similar to Flow-Based Generative Models, to enhance simulation fidelity and ensure numerical stability
- Increased model reliability through rigorous statistical validation on over 50,000 correlated data points, applying methods like AIC-based selection and chi-squared minimization with full covariance matrices.

2. **North Carolina State University** - Mathematical Physics

(Dec 2020 - Present)

- Implemented and managed Mathematica symbolic computation workflows on HPC clusters to analyze complex algebraic structures and symmetry constraints.

Publications

1. C.-R. Ji and **H. Ravikumar**, "*Interpolating conformal algebra in $(1+1)$ dimensions between the instant form and the light-front form of relativistic dynamics*," **Physical Review D** (2026). DOI: 10.1103/prlq-4j1l.

Technical Projects

1. **AI-DataScience-Lab: Full-Stack Data Product**

GitHub | Live App

- Prototyped and deployed a scalable full-stack data tool (Python, Flask, Azure) to deliver on-demand data-driven insights and actionable recommendations, integrating an OpenAI API to demonstrate automated data storytelling and data interpretation.
- Implemented an end-to-end statistical analysis pipeline using Python (Pandas) for data cleaning, statistical modeling (Scikit-learn) for regression, and Matplotlib for data visualization.

2. **Physics Based Monte Carlo Simulation**

GitHub

- Developed a Physics-Based Simulation from scratch to generate synthetic lattice gauge configurations using Monte Carlo methods on HPC clusters, validating the generated data against known analytical benchmarks.

3. Neural Network from Scratch with NumPy

GitHub

- Implemented and trained a neural network from scratch in NumPy for MNIST digit recognition, achieving 80% accuracy by building and tuning core components like backpropagation and activation functions

Technical Skills

Programming	Python, C++, CUDA, Bash, SQL, Lua, HTML/CSS, YAML
ML & APIs	Numba, TensorFlow, PyTorch, Scikit-learn, Pandas, cuFFT, cuDNN, Flask, FastAPI, RAG
Cloud & MLOps	Azure, AWS (Lambda, S3), CI/CD, Docker, Git, SLURM, Nsight
Methods & HPC	Parallel Computing (GPU, MPI), Numerical Methods (PDEs, Monte Carlo, Regression)

Education

PhD in Physics , New Mexico State University, USA	<i>Aug 2021 – Dec 2026 (expected)</i>
MS in Physics , New Mexico State University, USA	<i>Aug 2021 – May 2024</i>
MSc in Physics , National Institute of Technology Jalandhar, India	<i>July 2019 – May 2021</i>
BSc in Physics , Dr. N.G.P. Arts and Science College, India	<i>June 2015 – May 2018</i>

Certifications

- Getting Started with Accelerated Computing in CUDA C/C++ by NVIDIA
- Fundamentals of Accelerated Computing with CUDA Python by NVIDIA
- Advanced Learning Algorithms by DeepLearning.AI
- Supervised Machine Learning: Regression and Classification by DeepLearning.AI
- Google Advanced Data Analytics Professional Certificate

Awards

- **2025 NMC Collaboration Grant**, awarded by the New Mexico Consortium to conduct my independent research project in collaboration with scientists at Los Alamos National Laboratory
- **2023 George and Barbara Goedecke Physics Excellence Fund Scholarship**, awarded by the NMSU Physics Department

Selected Talks

- (Jun 3, 2025) "*First Principles Lattice QCD Calculations of n EDMs*", T-2 Seminar, Theoretical Division, **Los Alamos National Laboratory**, USA
- (May 16, 2024) "*Lattice QCD Calculations of x Dependence of Sivers TMD*", T-2 Seminar, Theoretical Division, **Los Alamos National Laboratory**, USA
- (June 15, 2023) "*Lattice QCD calculations of TMDs*", HUGS Student Seminar, **Thomas Jefferson National Accelerator Facility**, USA

Full list available at: hariprashad-ravikumar.github.io/talks

Volunteering

- **Vice President**, Physics Graduate Student Organization (NMSU) *Aug 2025 – Present*
Organized professional development events and served as the primary liaison between 40+ graduate students and faculty.

Relevant Graduate Coursework

- Quantum Computing, Advanced Computational Physics, Statistical Mechanics